

Geysering Generated by Large Air Pockets Released through Water-Filled Ventilation Shafts

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Abstract: One potential problem affecting below-grade storm-water storage tunnels is the occurrence of geysering, which is defined as the return of conveyed water to grade. Most investigations to date have linked this occurrence with inertial oscillation of the water within vertical shafts. Another mechanism that can lead to geysering is the release of air and water through ventilation towers. This study presents a systematic investigation on geysering caused by the release of large air pockets through partially water-filled ventilation towers. Parameters considered in the study included the water level in the ventilation tower, air-phase pressure head, and ventilation tower diameter. An important parameter in geysering was the diameter of the ventilation tower. A simplified numerical model was developed to simulate the experiments; it was able to reproduce the essential features of the experiments. DOI: 10.1061/(ASCE)HY.1943-7900.0000332. © 2011 American Society of Civil Engineers.

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Introduction

Large underground storage tunnel systems have been used in the past decades as a means to mitigate the impact of intense rain events in urban areas, allowing for runoff storage and conveyance while reducing the frequency of combined sewer overflow events. As the application of these tunnels become more common, operational problems related to rapid filling events became apparent. One of these problems is the return of storm water to grade, which is commonly called as *geysering*, although there does not seem to be consensus on a precise definition for this term.

There are at least two different mechanisms that could explain the return of storm water to grade. The first mechanism is based on the development of inertial oscillations of the water mass within the storm-water tunnels. Such pressure surges could cause the water in drop shafts to rise fairly quickly and reach grade, an event that could be considered to be a geyser. This possibility has been considered in the investigations presented by Guo and Song (1990, 1991).

The other mechanism that has been linked with the occurrence of geysers is the combined release of air and water through vertical shafts. The experimental investigations by Vasconcelos and Wright (2006) on rapid filling events showed several different mechanisms by which large air pockets may be entrapped in rapidly filling tunnels. These experiments also qualitatively identified at least three different mechanisms resulting in geysering through ventilation towers related to air and water release:

- Large air pocket escape through partially water-filled ventilation towers—this mechanism occurred when entrapped air pockets migrate along the tunnel crown where pressurized flow conditions exist. As the large air pockets arrive at a water-filled ventilation tower, air rises because of its buoyancy and pushes the water within the tower upward ahead of it, creating the geyser. Fig. 1 illustrates this mechanism;
- Release of an air/water mix through empty ventilation towers—the roller that characterizes the pressurized region of a pipe-filling bore can be viewed as a mixture of water and small air bubbles. As a result, the effective density of this mixture is smaller than the water density. When a pipe-filling bore reaches an empty ventilation tower, the air/water mixture rises as a surge depending on the pressure head behind the pipe-filling bore and the mixture density. Smaller densities favor the rapid filling of the ventilation tower, and in some cases the spilling level may be reached, generating geysering at that location;
- Small air-pocket release—this is caused by the rapid filling of a vertical shaft caused by the passage of a pipe-filling bore. In this

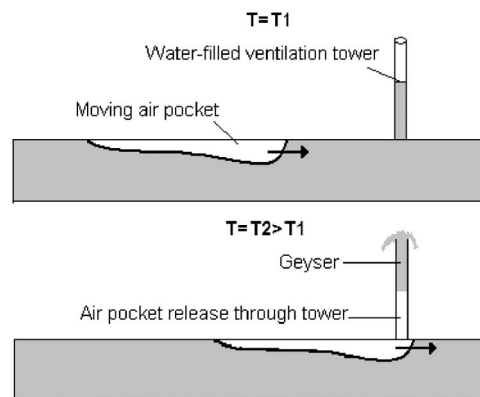


Fig. 1. Geysering generated by release of large air pockets through water-filled ventilation towers

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case, an air pocket that fills the cross section of the ventilation tower is entrapped between two layers of water. Rapid pressurization of the air pocket can result from the relative motion of the upper and lower water layers, compressing the pocket. In such conditions, the air pocket effectively acts like a spring, increasing the upward velocity of the upper water layer and strengthening the intensity of the geyser. This springlike effect of entrapped air pockets has been studied by Izquierdo et al. (1999) in the investigation of the startup of water mains.

Geysers have been reported in a number of storm water and combined sewer systems. Limited understanding of the relevant phenomena involved and the lack of adequate field measurements allows for the possibility that the effects of any or all of the conditions previously described have been called geysers. There has been an effort to instrument a storm-water tunnel system in Minneapolis, Minnesota, where numerous geyser events have been experienced. On the basis of the analysis of data provided for a single storm event, Wright et al. (2011) conclude that the formation of geysers in that system was related to the interaction of large entrapped air pockets at the monitored manhole. It is suspected that this is a major cause of the large explosive geysers that shoot jets of water and air tens of meters into the air through manholes or ventilation shafts. Thus, to minimize geysering impacts, it is important that the design of drop shafts and ventilation points consider the possibility of combined air and water release.

Objectives

The main objective of the present work was to investigate the geyser phenomenon caused by the release of large air pockets through ventilation towers. A scale-model apparatus that included the main features of a ventilation tower in a storm-water storage tunnel was constructed. Selected parameters were systematically varied to assess their effect while pressure and free-surface and air-water interface displacement data were collected. The experimental data collected were subsequently compared with the predictions of a numerical model designed to simulate the essential features of this vertical two-phase flow. The specific objectives of this investigation were as follows:

- To measure the rising velocity of the air pockets in the ventilation towers and compare the measurements with the predictions of a rigid column, vertical, two-phase flow model;
- To determine the characteristics of ventilation tower geometry that lead to significant water rise;

- To measure the depressurization of the system during the rising of the air pocket through the ventilation tower; and
- To develop a numerical model capable of replicating the experimental results.

Experimental Program

The experimental apparatus used in the experiments was installed on a horizontal slope, as represented in Fig. 2. The apparatus consisted of the following components:

- Acrylic upstream pipe with length of 0.546 m and diameter of $D = 0.094$ m;
- Air pump connected to the upstream pipe;
- Differential manometer connected to the upstream pipe;
- Butterfly valve ($D = 0.102$ m) connecting the upstream pipe and the middle pipe;
- Acrylic middle pipe with length of 2.97 m and diameter of $D = 0.094$ m. This pipe connected the butterfly valve to the PVC coupling in which the ventilation towers were installed;
- PVC coupling connecting the middle pipe, the ventilation tower, and the downstream pipe;
- Ventilation tower with variable diameter and length of $L = 0.610$ m;
- Scale attached to the ventilation tower length;
- Acrylic downstream pipe with length of 0.490 m and diameter of $D = 0.094$ m connected upstream to the PVC coupling and closed downstream;
- Piezo-resistive pressure transducer (8510B-1; Endevco, San Juan Capistrano, CA) (gauge pressure of 6.9 kPa) located 1.07 m downstream from the butterfly valve;
- Data acquisition board (DAQ-Pad MIO-16XE-50; National Instruments, Austin, TX); and
- Digital camcorder (with speed of 30 frames per second) recording water level variation in the ventilation tower.

The experimental procedure is schematized in Fig. 3 and can be described as follows:

1. With the butterfly valve initially closed, the upstream pipe was emptied, and pressurized air was injected into the pipe with an air pump. The desired pressure was regulated with the aid of a differential manometer (precision of 0.031 m) connected to this pipe;
2. Downstream of the valve, the pipe was entirely filled with water carefully to eliminate any air pockets present at the top of the pipe. The pipe was filled with water to a

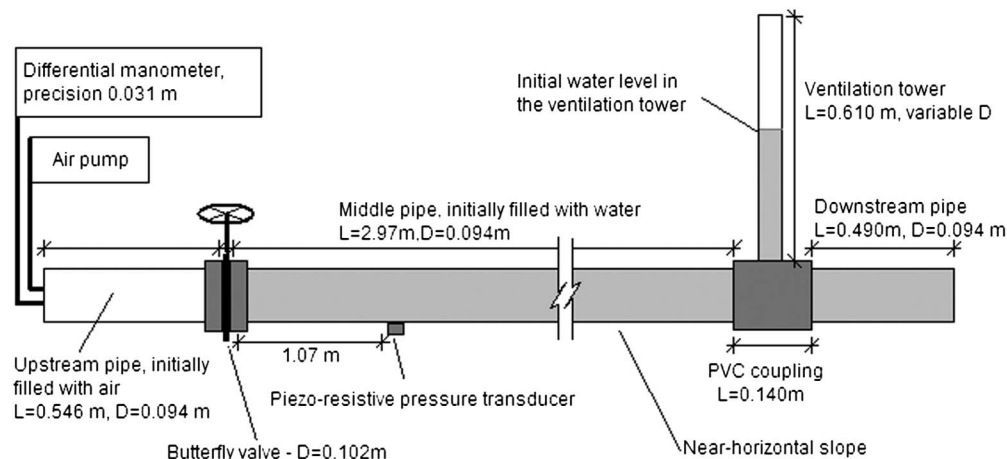


Fig. 2. Experimental apparatus used in experiments

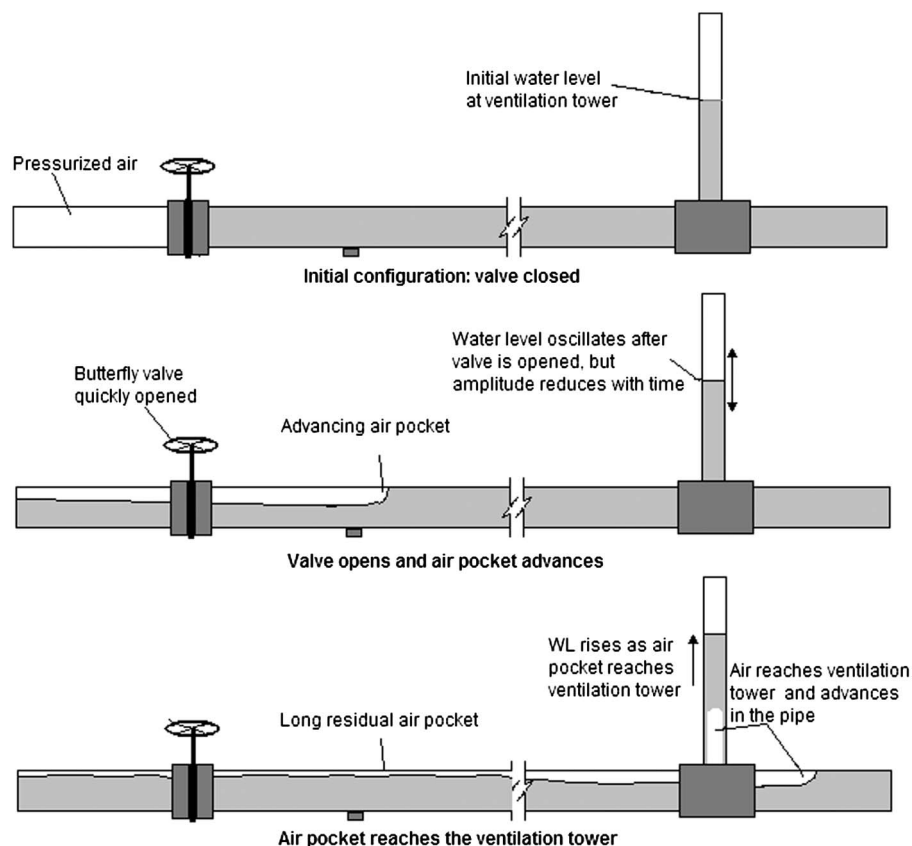


Fig. 3. Flow evolution during geysering experiments

predetermined level in the ventilation tower, which was measured with a scale attached to the tower;

3. After the air-phase pressure and the ventilation tower water level were at the desired levels, the camcorder and the pressure transducer were started;
4. The butterfly valve was manually opened (less than 1 s), and an air pocket started to advance toward the ventilation tower while water flowed underneath the pocket toward the upstream pipe. It was observed that the water level at the ventilation tower oscillated immediately after the opening of the valve because of nonequilibrium in pressure across the valve and that the amplitude of this oscillation reduced with time;
5. As the air pocket reached the location of the ventilation tower, a fraction of it moved upward. The air-water interface that rose in the vertical tower had a shape that resembled a Taylor bubble (Wallis 1969). However, unlike regular Taylor bubbles, there was not a trailing air-water interface, and the rising air pocket kept contact with the air that remained in the horizontal pipe.
6. The water level in the ventilation tower also rose ahead of the rising air-water interface. The water level at the ventilation tower and the location of the air-water interface were measured from the videos by using the scale attached to the ventilation tower. Because of the water film that flowed downward around the perimeter of the air pocket as it moved upward, the velocity of the free surface in the ventilation tower was smaller than the velocity of the rising air pocket;
7. In some cases, water would spill from the top of the ventilation tower, constituting what was defined as a *geysering occurrence*;
8. The camera and the pressure transducers would be stopped after 60 s. The initial and final water levels measured in the middle pipe were used for the subsequent calibration of the

pressure transducers. Every experimental run was repeated at least three times to ensure consistency of the experimental results.

Three experimental variables were varied to create different configurations in which large air pockets would be expelled through ventilation towers in storage tunnels. The first variable was the initial air phase pressure in the upstream pipe, which was chosen to assess whether stronger air pressurization would favor the occurrence of geysering. The second variable was the initial water level in the ventilation tower, which was selected to assess whether a higher initial water level water at the tower would favor geysering occurrence. The last experimental variable was the diameter of the ventilation tower, which was selected to assess whether a larger ventilation tower diameter would help prevent geysering. Thirty-six different configurations were tested, as presented in Table 1.

Another variable considered as potentially important was the initial volume of air in the system, which was regulated by placing a predetermined volume of water in the upstream pipe before the start of the experimental run. The goal in reducing the amount of air was to check whether smaller air pockets could create different types of shapes of the air-water interface as it moved upward in the ventilation tower. However, even when the initial volume of

Table 1. Range of Experimental Variables Considered in Experiments of Geysering Caused by Release of Large Air Pockets

Experimental variable	Considered ranges
Air-phase initial pressure head (m)	0.305, 0.610, and 0.915
Initial water level at ventilation tower (m)	0.254, 0.356, and 0.457
Ventilation tower diameter (mm)	57.1, 44.4, 25.4, and 12.7

air was decreased to 25% of the volume of the upstream cavity, the shape of the upward moving pocket at the ventilation towers with smaller diameters was unchanged. Thus, this variable was dropped from the experimental protocol. Also, a shorter ventilation tower length ($L = 0.305$ m) was initially used in the experimental program. However, because geysering occurrence with the short tower was much more frequent, whereas it was desired to study some cases when geysering did not occur, that configuration was eventually disregarded.

Experimental Results

After the completion of the experimental runs, the recorded movies were uploaded from the camcorder into a computer. The water level in the ventilation tower measured with the scale was determined from each video frame. As the air pocket rose in the tower, the location of the top of the rounded air-water interface measured with the scale was also recorded. Finally, the results obtained with the pressure transducers after calibration were included in spreadsheets for further analysis. The small discrepancies in the timing between the repetitions, especially noticed in the pressure transducer output, are attributed to the manual operation of the butterfly valve, resulting in slight differences of the opening times.

The pressure transducer measurements (H^*), the free-surface level (Y_{fs}^*) and air-water interface location (Y_{int}^*) were normalized by the ventilation tower length L . The initial value of Y_{fs}^* depended on the initial water level in the ventilation tower, and it increased as the air pocket reached the bottom of the ventilation tower and started moving upward. The initial value of Y_{int}^* was zero and increased as the air pocket moved upward. When $Y_{int}^* = Y_{fs}^*$, the upper interface of the rising air pocket reached the same location as the free surface, creating a free path for air to escape to the atmosphere. If $Y_{fs}^* = 1$ before $Y_{int}^* = Y_{fs}^*$, then water would spill from the top of the ventilation tower, characterizing the geyser event.

The results were grouped by the nondimensional ventilation tower diameter D_t^* , which was normalized by the horizontal pipe diameter D . The results were also grouped by the type of measurement. Time was normalized by dividing it by $L/\sqrt{gD_t}$. As Table 1 indicates, for each one of the four ventilation tower diameters, there were nine different combinations of initial air-phase pressure and initial water level at the ventilation tower. Also, for each one of the 36 experimental conditions, five different experimental variables

were measured or computed, resulting in 180 graphs, as presented in Vasconcelos (2005). Each graph contains the results of the three repetitions of the experimental conditions. Because of the large amount of data, results are presented only for the largest and smallest ventilation towers used in the investigation ($D_t^* = 0.607$ and $D_t^* = 0.135$).

An interesting observation in the experiments was the asymmetric nature of the vertical air pocket in the initial stages of the rising motion. Fig. 4 shows five different snapshots of an experimental run with $D_t^* = 0.607$, initial water level of 0.356 m, and air pressure of 0.305 m. The time separation between successive images is approximately 0.14 s, and the sequence covers the advance of the air pocket from 0.05–0.26 m above the horizontal pipe crown. In this figure, the horizontal air-pocket motion before the arrival at the ventilation tower was from left to right. The initial shape of the rising pocket is asymmetrical because of air entering first from the left side of the tower, and as it advances up the tower, the shape of the air pocket gradually evolves into a symmetrical front that resembles the leading edge of a Taylor bubble.

Pressure Measurements

Strong pressure fluctuations were observed in many of the experiments following the opening of the butterfly valve. These fluctuations were caused by the initial pressure difference between the air and water phases before the opening of the valve. The pressure fluctuations were also noticed in the free-surface level at the ventilation tower, causing it to oscillate immediately after the opening of the valve. The amplitude of these fluctuations decreased quickly after the opening of the valve; however, in a few cases, there were still some residual fluctuations at the time when the air pocket reached the ventilation tower, particularly for the larger-diameter towers. This was somewhat undesirable because the fluctuations affected the reading of the free-surface level. An air pressure head greater than the initial pressure head in the ventilation tower caused the water level to increase above its initial level after the butterfly valve was opened.

The rise of large air pockets in vertical towers can be viewed as a transient, two-phase flow problem comprising the upward motion of an air pocket in the core of the tower and the downward flow of a thin layer of water (called *film flow*) attached to the perimeter of the tower around the air pocket. In the experiments, when the air pocket reached the ventilation tower and started the upward motion, the film flow caused a decrease in the length of the water column

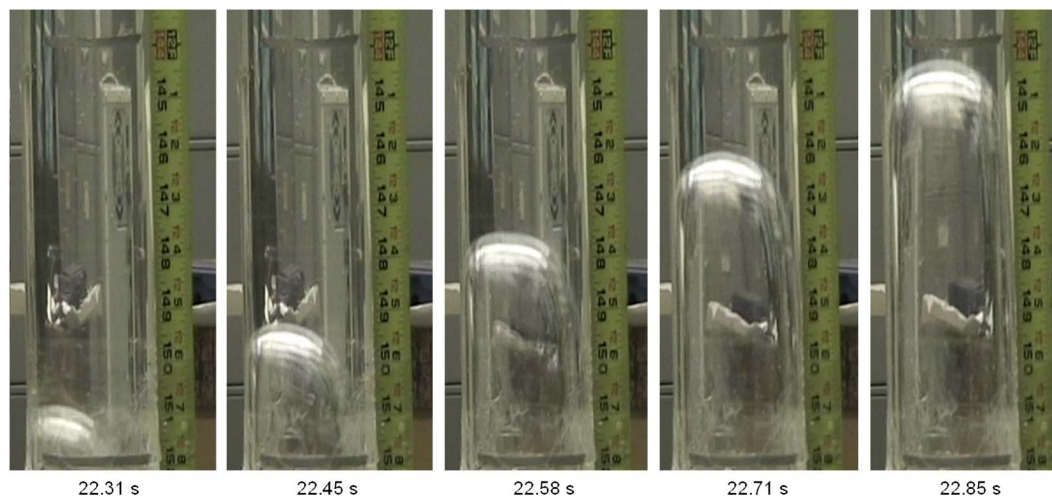


Fig. 4. Rising motion of air pocket; horizontal motion of air pocket in pipeline is from left to right, the side scale is in inches, and $D_t^* = 0.607$

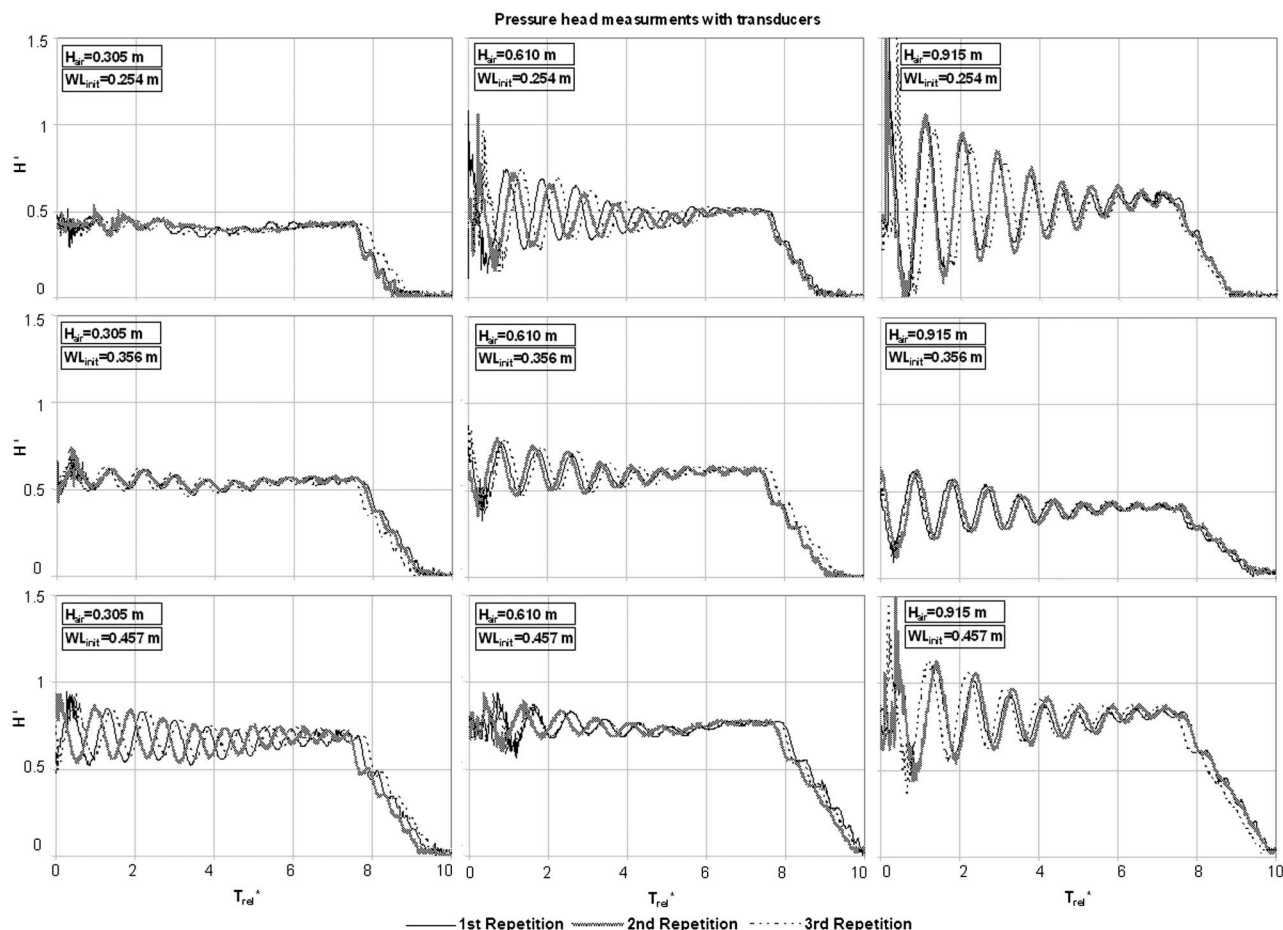


Fig. 5. Normalized pressure head measurements; $D_t^* = 0.607$

in the tower. This decrease in turn results in the decrease of the measured pressures, as demonstrated in Fig. 5 for $T_{ref}^* > 7$. The pressure decreases up to the moment when the pocket surfaces and the air pressure becomes atmospheric. The manual opening of the valve explains the small differences in the time when the pressure started to decrease among the three repetitions.

Distinct pressure oscillations are observed for the towers with $D_t^* = 0.270$ and greater, and the free-surface level exhibits this same general behavior. On the other hand, the pressure behavior was different for the smallest tower ($D_t^* = 0.135$), as shown in Fig. 6. Instead of oscillations, there was a monotonic decrease in pressure that became more rapid with time.

Measurements of Free-Surface Level and Air-Water Interface Levels

The measurements of the free-surface level were significantly influenced by the tower diameter. Generally, the larger tower diameters were associated with smaller variations in the free-surface level as the air pocket rose in the tower. The increase in the free-surface level for the cases when $D_t^* = 0.607$ was generally less than 10% of the tower length L , as presented in Fig. 7. For the cases when $D_t^* = 0.472$, the variation approached 20% of L in most cases. For the cases when $D_t^* = 0.270$, the water level variation could be as large as 40%, resulting in geysering in some of the runs. For the cases when $D_t^* = 0.135$, the free-surface level variation always generated geysers. A slight oscillation pattern in the rise of the free surface is noticeable. The amplitudes of these oscillations decrease for smaller tower diameters. The exception is the

case when $D_t^* = 0.135$, as there was a monotonic and accelerated upward movement of the free-surface level. The results from these measurements are presented in Fig. 8.

The air-water interface level increased more rapidly than the free-surface level as the water volume above the advancing interface decreased because of the film flow. For the larger-diameter towers, the air pocket surfaced without generating a geyser. Oscillations in the free surface and air-water interface level were also observed for the three larger-diameter towers. The results for the $D_t^* = 0.135$ case indicated an acceleration of the air-water interface advance. After geysering started, there was further acceleration in the upward motion of the air-water interface.

Effects of the initial water level in the ventilation tower and the initial air-phase pressure were minor in the experiments. Basically, these variables determined the starting point for the free-surface level as it increases because of the air-pocket rise. If this initial water level was closer to the spilling point, geysering was more likely to occur. Also, as in the case of the pressure oscillations following the butterfly valve opening, a larger discrepancy in the initial pressure between the phases caused larger oscillations in the free-surface level upon opening of the valve.

Free Surface and Air-Water Interface Velocities

The rising velocity of the free surface was obtained indirectly by measuring the recorded advance of the interfaces at regular intervals, usually around 0.20 s. In general, both the free-surface level and the air-water interface velocities decreased with increasing tower diameters. For the three larger-diameter towers, the averaged

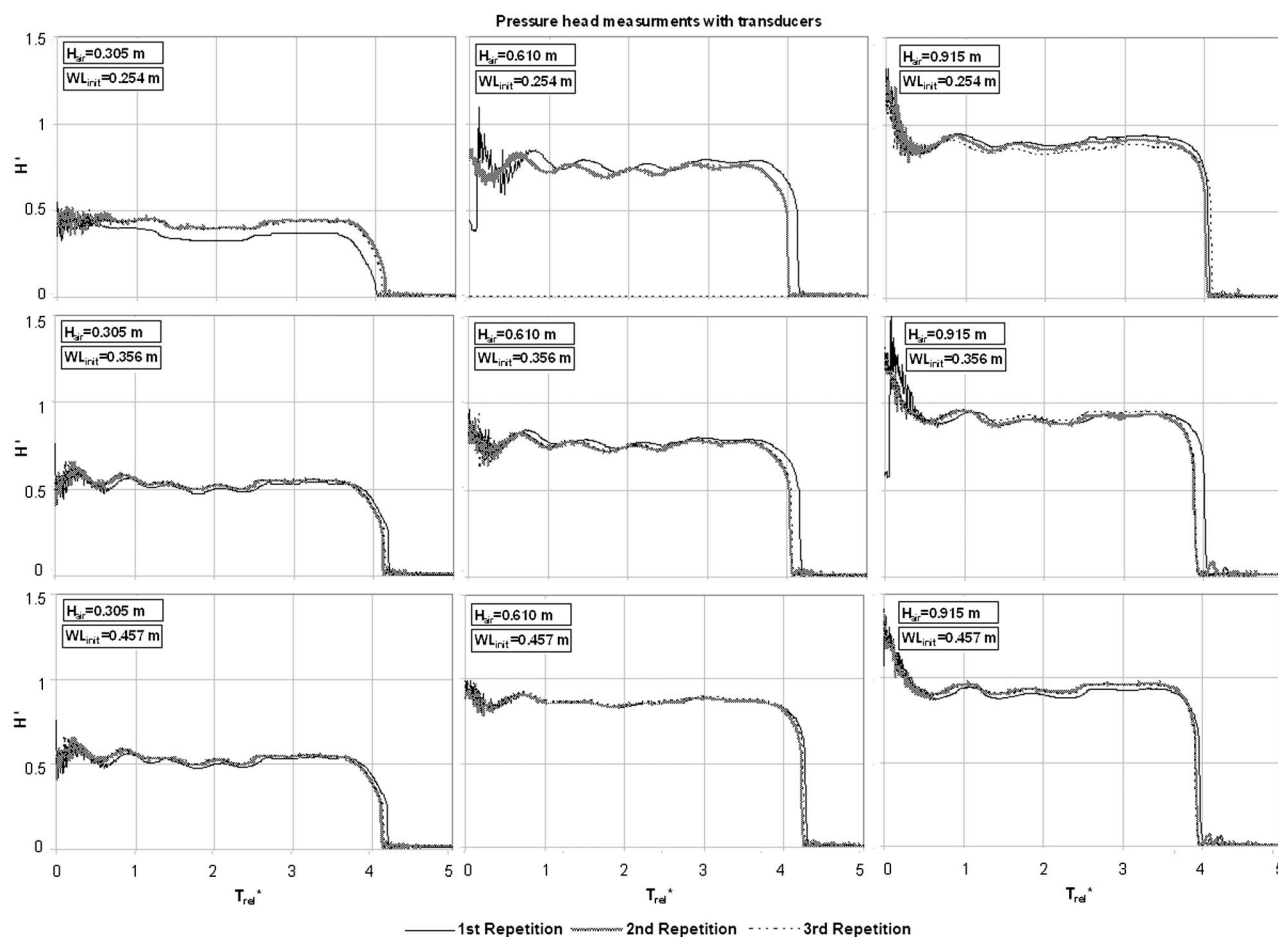


Fig. 6. Normalized pressure head measurements; $D_t^* = 0.135$

values for the free-surface velocity for each tower are presented in Table 2 and are normalized by $\sqrt{gD_t}$. Because of the oscillating behavior of the velocity measurements for the three larger tower diameters, the average velocity shown in Table 2 gives only a general idea of the upward air-pocket velocity. However, this observation is not valid for the velocity measurements for $D_t^* = 0.135$ because of the absence of velocity oscillations. The following are some observations from the velocity analyses:

- For larger D_t^* , the free-surface level velocity is much smaller than the air-water interface velocity. As D_t^* decreases, the free-surface level and the air-water interface velocity increase.
- The difference between the free-surface level velocity and the air-water interface velocity tend to increase as D_t^* decreases. Considering that the ventilation tower length was fixed for all experimental runs, conditions with larger differences are associated to geysering occurrence.
- The velocity measurements for the three larger D_t^* indicate the oscillation pattern in the velocities; for the smallest D_t^* there is a monotonic rise in both interface velocities.
- The measurements of the air-water interface velocity for $D_t^* = 0.135$ indicate further acceleration after the spilling of water at the top of the tower. The final nondimensional velocity of the air-water interface was often three times larger than the averaged V_{int}^* shown in Table 2.
- The air-water interface velocity was larger than the predictions of the vertical slug-flow theory for the problem of emptying a vertical pipe (Wallis 1969).

The classical theoretical framework of slug flows cannot be applied to this problem because there is an imbalance between

the volume displaced by the air pocket and the film flow, as indicated by the nonzero free-surface level velocity. Should the initial air volume in the horizontal pipe be much smaller, the upward moving air pocket would eventually have a trailing edge. In this case, drag forces would appear, changing the flow dynamics. It is speculated that a consequence of this additional force would be the deceleration of the pocket, resulting in velocities that would resemble the theoretical predictions of slug-flow theory.

The closest theoretical framework for the aforementioned problem is based on the application of emptying a vertical pipe addressed by Davies and Taylor (1950). In that study, the two-phase flow is associated with one upward-moving air cavity and the water that flows downward in the film attached to the walls. When the upward flow is established in the vertical pipe, the volume of the water displaced by the cavity is balanced by the film flow, and the free-surface level does not vary with time. That study demonstrates that the rising velocity of the air pocket scales with $\sqrt{gD_t}$.

However, unlike the pipe-emptying problem, the entrapped air pocket in the experiments is generally above atmospheric pressure. The presence of this additional air-phase pressure force can alter the balance between the buoyancy, inertia, and surface tension forces. Instead of a pipe being emptied at the bottom, the problem can be viewed as a pressurized air pocket that pushes a vertical water slug upward—a water slug that decreases in volume over time because of the film flow. This latter concept was considered in the derivation of a numerical model that aimed to reproduce the observations of these experiments.

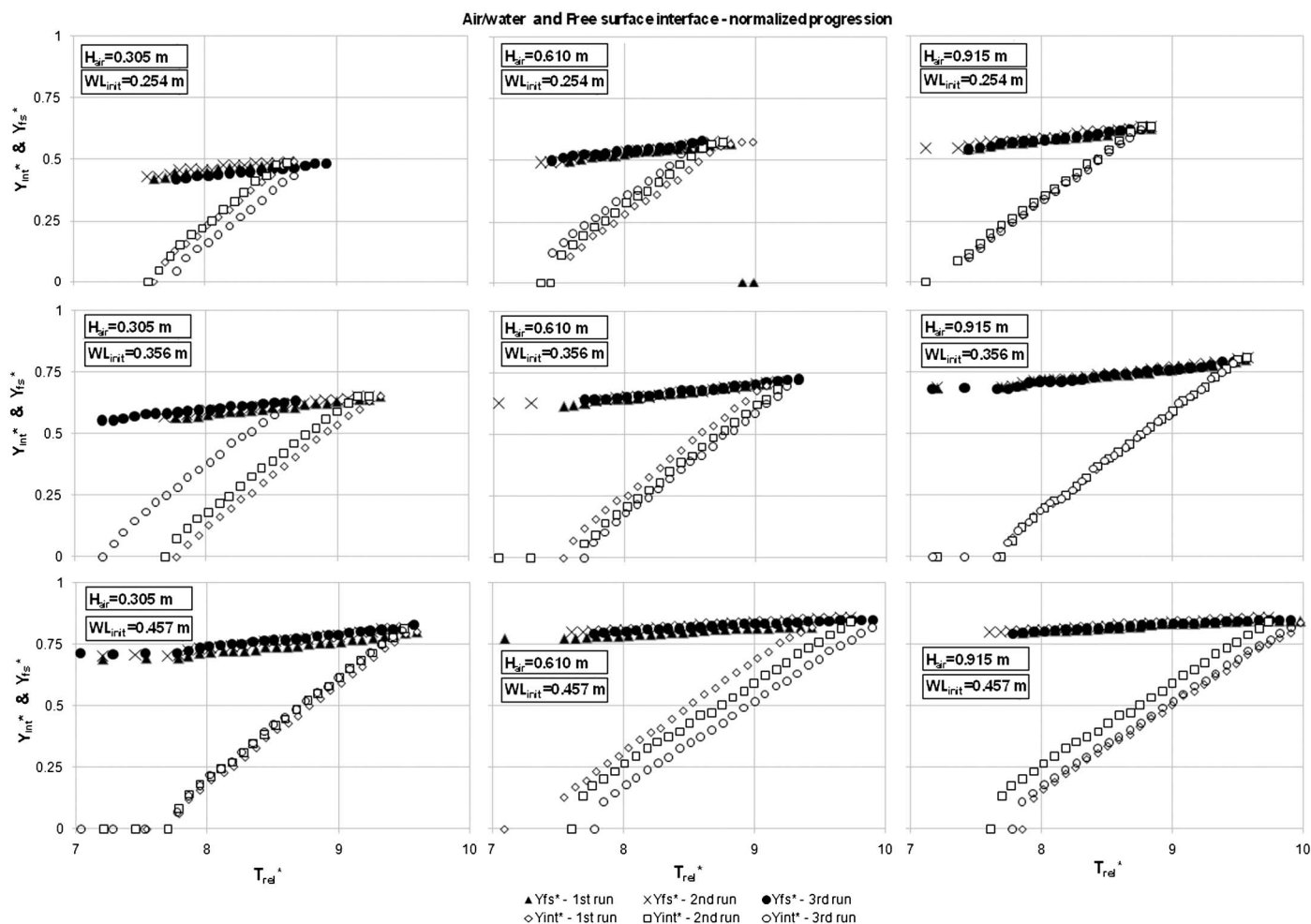


Fig. 7. Normalized free-surface level and air-water interface y-coordinates; $D_t^* = 0.607$

Numerical Model Approach

A numerical model was developed to simulate the measured results. This model is a relatively simple approach to describe the vertical two-phase flow of the rise of entrapped air pockets in water-filled ventilation towers. The model approach does not include the horizontal motion of the air pocket before it arrives at the ventilation tower (see Fig. 2).

The basic idea of the model is to describe the problem as a combination of three different physical mechanisms. The first one is the rise of a water column caused by the imbalance between the internal air pressure and the weight of the column. The second is the film flow, which decreases the volume of the water column and decreases the volume of air gained by the upward motion of the water column. The third and last mechanism is the air-phase compression/expansion process, which is controlled by the upward motion of the water column and the film flow.

The proposed model does not describe flow details such as the velocity distribution in the vicinity of the air-water interface. Key assumptions were introduced in the model development that allowed for a simple framework based on a system of ordinary differential equations detailed in the following.

The first assumption is that the problem can be analyzed in a one-dimensional framework. This assumption is probably adequate as long as Y_{int}/D_t is large enough. However, the asymmetry of the air pocket in the initial moments of the upward motion (illustrated

in Fig. 4) could pose some limitations to this assumption, at least for the initial period of the numerical simulation.

The second assumption is that the problem can be viewed as the upward motion of a water column driven by the air-phase pressure, with the requirement that the water column volume decrease because of the film flow. It is further assumed that the initial air-water interface velocity equals the rising velocity of a Taylor bubble; thus, $V_{int,t=0} = U_\infty = 0.345\sqrt{gD_t}$, where g = gravitational acceleration. This requirement of instant initial air-pocket velocity is corroborated by experiments performed by Baumbach et al. (2005). The authors point out that air pockets rising in static water experience an acceleration as large as two times the gravitational acceleration in the initial stages of the motion when time $t < \sqrt{D_t/g}$. For larger times, the rising velocity approached the predicted theoretical values. For the case of the presented experiments, this initial stage based on their criterion ranges between 0.035 and 0.076 s, which is much smaller than the overall rising time of the air pocket.

The third assumption is that the decrease of the water column height caused by the film flow is independent of the upward motion caused by the entrapped air-pocket pressure. Two simplifications are derived from this assumption. The first simplification is that flood instabilities (Guedes de Carvalho et al. 2000) were not considered in the calculations. Although this is true for the scale in which the experiments were conducted, one anticipates that it would be an important flow feature for full-scale applications. The second simplification is that the film flow was calculated as

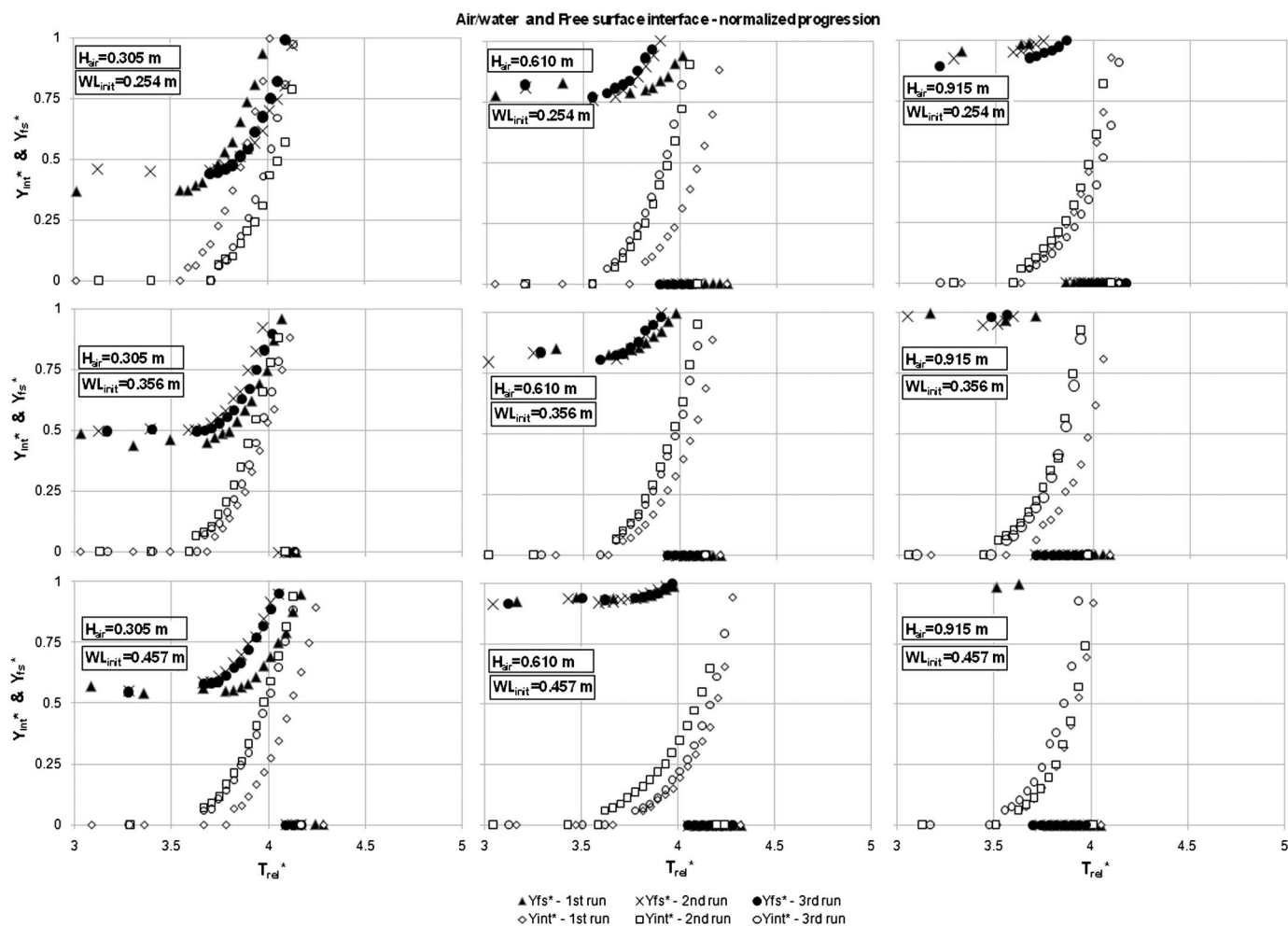


Fig. 8. Normalized free-surface level and air-water interface y-coordinates; $D_t^* = 0.135$

in the case of a vertical pipe-emptying problem. At the first time step, before the air-phase pressure induces the motion of the water column, the net volume flow in the ventilation tower is zero, such as in the case of the emptying of a vertical pipe. Thus, the laminar film flow Q_f around a vertically rising air pocket is matched by the water displaced by the pocket $Q_{\text{displaced}}$:

$$Q_{\text{displaced}} = Q_f \quad (1)$$

$$\frac{\pi}{4} U_{\infty} (D_t - 2\delta)^2 = \frac{\pi}{3} \frac{g D_t (\rho_w - \rho_a) \delta^3}{\mu_w} \quad (2)$$

in which δ = film flow thickness; ρ_w , ρ_a = water and air densities, respectively; and μ_w = dynamic viscosity of water. Eq. (2) is similar to the equation proposed by Batchelor (1967) for laminar film, except for the addition of the term $\pi U_{\infty} \delta^2$ on the left hand side

of the equation. Eq. (2) is solved to determine the film thickness that is used throughout the computation. The calculation of the free-surface level depends both on the vertical motion of the water column and the film flow that decreases the length of this column.

The final assumption is that air phase undergoes an isentropic expansion/compression process. The numerical model and the experimental results are compared for the cases when the initial $H_{\text{air}} = 0.305$ m and the initial $Y_{\text{int}} = 0.254$ m to minimize the pressure oscillations caused by the initial discrepancy between the air- and water-phase pressures. Also, the variations in the air-phase volume after the butterfly valve is opened are neglected. Thus, it is assumed that the initial air volume in the system is equal to the volume of the upstream pipe behind the butterfly valve (see Fig. 2). The volume of the air phase increases as the air pocket rises in the ventilation tower, but the film flow diminishes this expansion to some extent.

With regard to Eq. (2), it is shown in Batchelor (1967) that laminar film thickness scales with $D_t^{1/2}$. Considering the ventilation tower perimeter, the laminar film flow scales with $D_t^{5/2}$. As for the left-hand side of Eq. (2), because the term U_{∞} scales with $D_t^{1/2}$, the overall left-hand side scales with $D_t^{5/2}$. However, larger film thickness tends to decrease the magnitude of the left-hand side of Eq. (2) (through the $D_t - 2\delta$ term) while increasing the right-hand side. One then anticipates that larger ventilation tower diameters should decrease the likelihood of geysering occurrence

Table 2. Averaged Upward Nondimensional Velocities for Different Tower Geometries

D_t^*	Average V_{fs}^*	Average V_{int}^*
0.607	0.048	0.39
0.472	0.088	0.45
0.270	0.30	0.79
0.135	0.44	1.43

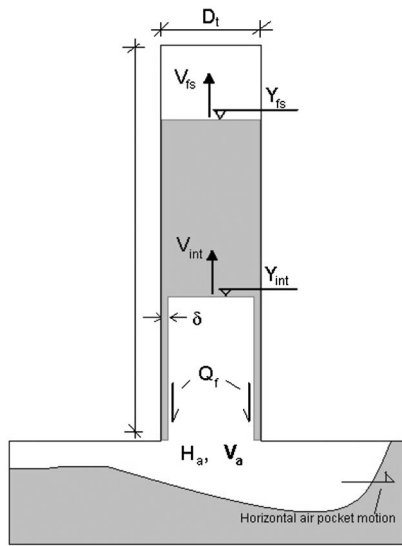


Fig. 9. Schematic of variables considered in geyser model

caused by large air pockets release, which is in agreement with the experimental results.

Fig. 9 indicates the variables included in the numerical model. With the initial estimation of the air-water interface velocity $V_{int,t=0} = U_\infty$ and the thickness of the film flow δ , the following system of ordinary differential equations was constructed:

- Air-water interface velocity

$$\frac{dV_{int}}{dt} = \frac{g}{Y_{fs} - Y_{int}} \left[\frac{Q_f V_{int}}{gA} + H_a - (Y_{fs} - Y_{int}) - f \frac{(Y_{fs} - Y_{int}) |V_{int}| V_{int}}{D_t} \right] \quad (3)$$

- Air-water interface location

$$\frac{dY_{int}}{dt} = V_{int} \quad (4)$$

- Air volume variation

$$\frac{dV_a}{dt} = \frac{\pi}{4} (D_t - 2\delta)^2 V_{int} - Q_f \quad (5)$$

- Air-phase pressure head

$$\frac{dH_a}{dt} = \frac{1}{\rho_w g} \left[\frac{\gamma C_a}{V_a} \left(\frac{\rho_w g (H_a + H_{atm})}{C_a} \right)^{\gamma/(1-\gamma)} \times \left(\frac{\rho_w g (H_a + H_{atm})}{C_a} \right)^{1/\gamma} \frac{dV_a}{dt} \right] \quad (6)$$

- Free-surface level coordinate and velocity

$$\frac{dY_{fs}}{dt} = \frac{1}{A} \frac{dV_a}{dt} = V_{fs} \quad (7)$$

In the preceding equations, A = cross-sectional area of the ventilation tower; V_a = volume of the air phase; H_a = air-phase pressure head; H_{atm} = atmospheric pressure head (10.33 m at sea level); γ = polytropic coefficient assumed as 1.4; and C_a = constant equal to P_{a0}/ρ_a , with P_{a0} = initial absolute pressure in the upstream pipe before the butterfly valve opening. The approach to calculate the air-phase pressure follows Li and McCorquodale (1999). The friction factor was calculated explicitly following an expression presented by Arai and Yamamoto (2003), and the material relative roughness (ϵ/D_t) was assumed as 0.0015.

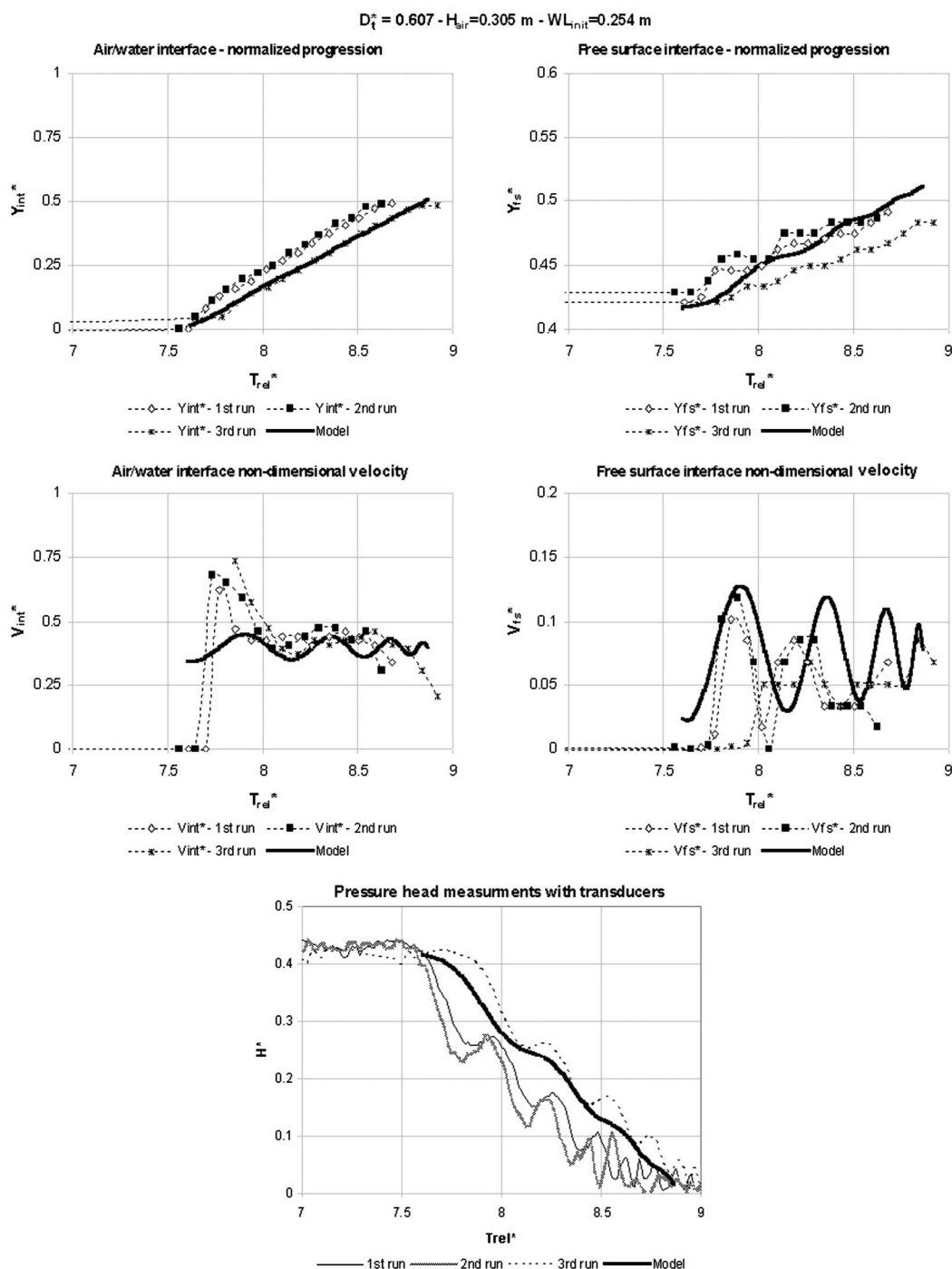
The proposed numerical model solves Eqs. (3)–(7) at each time step by using a Runge-Kutta fourth-order method, as described by Press et al. (1989). The time-step size used in the computations was 10^{-5} s. The program interrupts the calculations if one of two conditions happens: either $Y_{int}^* = Y_{fs}^*$, meaning that the air pocket has surfaced without the occurrence of geysering, or $Y_{int}^* = 0.99$, corresponding to the case when the air pocket has effectively reached the top of the ventilation tower and geysering was generated by the air-pocket motion. The calculations in the latter case need to be interrupted as Y_{int}^* reaches 0.99 to avoid division by zero in Eq. (3). In Eq. (3), the contribution of the film flow to the momentum balance was not included. This is based on the assumption of laminar flow in which shear forces and gravity in the film are balanced, resulting in no acceleration for the film flow. The fact that the proposed numerical model assumes that the film flow is a laminar flow restricts the applicability of the proposed model to small-scale applications, such as the ones that presented in the experimental studies.

Assessment of Numerical Model

The accuracy of the proposed numerical model for the geysering generated by the release of large air pockets is assessed by comparing the model results with the experimental data collected in the experiments. The comparison was performed for each of the ventilation towers used in the experiments and for the cases when the initial $H_{air} = 0.305$ m and the initial $Y_{int} = 0.254$ m. This choice of initial pressure heads results in reduced pressure oscillations compared with those observed when the pressure difference between the phases was higher.

Fig. 10 presents the comparison for the case when $D_t^* = 0.607$. The agreement between the numerical model and the experiments with regard to the prediction of the variation of the free-surface level and air-water interface is generally good, capturing the essential features of the flow. With regard to the interface velocities and pressure predictions, the agreement is not as good, but essential flow features were captured. The pressure behavior was also fairly well predicted, presenting the same decrease trend. The period and amplitude of the oscillations do not match the experimental observations. This could be attributed to the simplifying assumptions used in the model, particularly that the vertical acceleration of the water column caused by the air pocket has no influence on the film flow. Also, the assumption of this problem being one-dimensional may be a limiting factor in the model accuracy, particularly for the case of towers with large D_t/L ratios.

Fig. 11 presents the comparison for the case when $D_t^* = 0.135$. The model correctly predicts the experimental observations of geysering occurrence. Instead of the oscillation pattern for the interface trajectory and velocities, the numerical model predictions present a monotonic variation of the variables, agreeing with the experimental results. An interesting characteristic of the numerical model predictions is the effect of the geysering initiation on the results of the nondimensional air-water interface velocity. Upon initiation of geysering, the predicted increase in V_{int}^* becomes more rapid. This is caused by the additional mass loss at the top of the tower caused by the spilling, which alters the momentum balance and generates further acceleration of the water column. This is captured by the numerical model, as V_{int}^* accelerates after $T_{ref}^* > 4.1$. The model cannot replicate the sharp decrease of the pressure head, with the predicted final pressure head of $H^* = 0.25$. This could be linked to the fact that the model stops its computation slightly before Y_{int} reaches Y_{fs} to avoid division by zero on Eq. (3).



Other conditions not included in the experimental program were simulated with the numerical model to predict the maximum free-surface level caused by the air-pocket rise for different values of the initial water level in the tower. In these numerical simulations, the ventilation tower length was increased to ensure no spilling from the top of the tower. The tower diameter was the same as that used in the experimental program, and five initial water level were considered.

The results of the 20 simulated cases are presented in Fig. 12, with the horizontal axis presenting the initial water level (normalized by the tower diameter), and the vertical axis presenting the final free-surface level in the tower (normalized by the initial free-surface level).

The results indicate that $Y_{fs}/Y_{fs,0}$ increased for smaller D_t , which is in agreement with the experimental results. The ratio between the final free-surface level to its initial level $Y_{fs}/Y_{fs,0}$ was fairly constant, with a mild decline as $Y_{fs,0}$ increased. It is important that these results are viewed with caution, considering that there is no experimental confirmation (except for the tested cases) nor the scale in which those numerical runs were performed account for all interactions between air and water phases in a prototype scale.

Discussion

The results indicate that geysers were less likely to occur for larger-diameter ventilation towers. This is probably related to the fact that

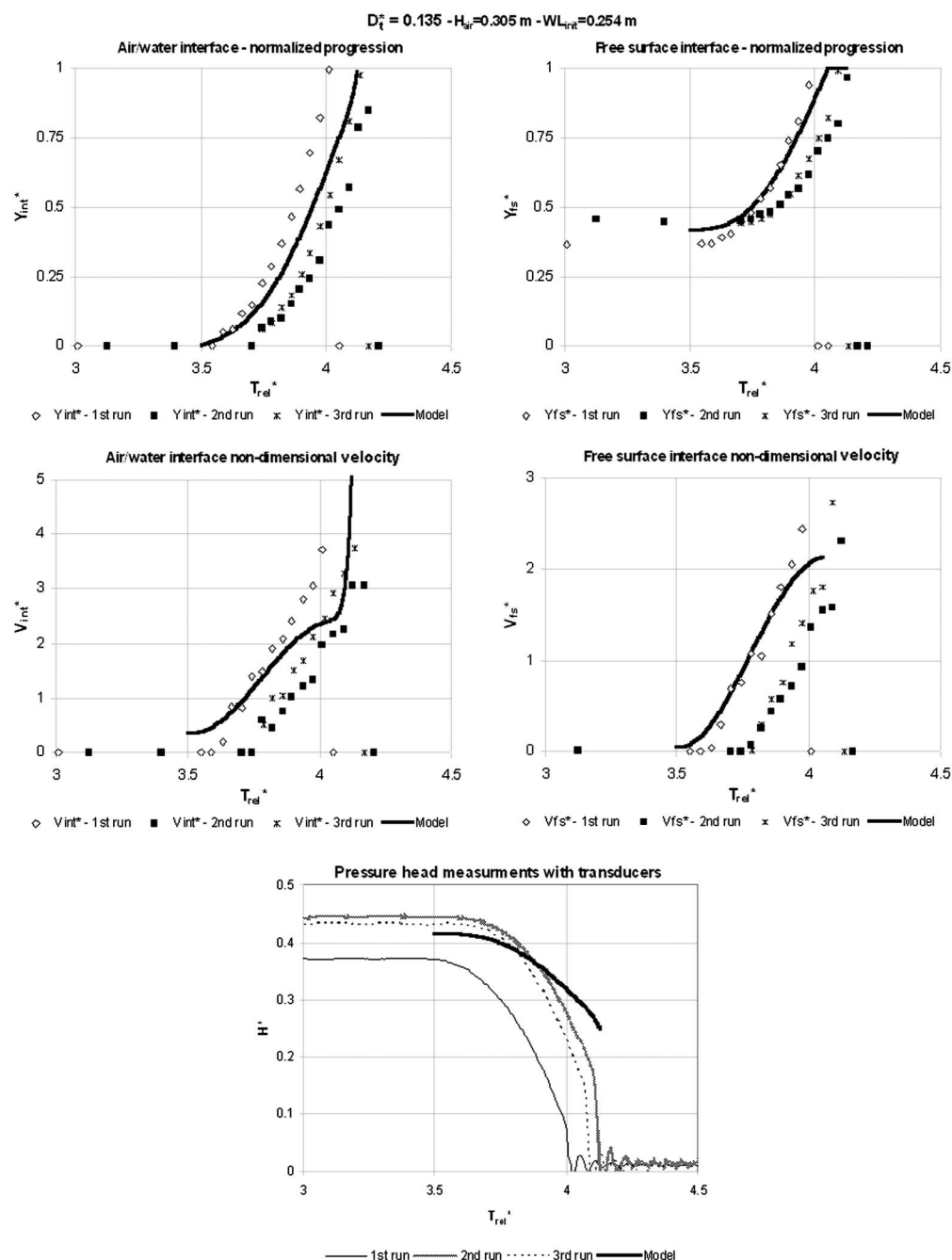


Fig. 11. Comparison between numerical model and experimental data; $D_t^* = 0.135$

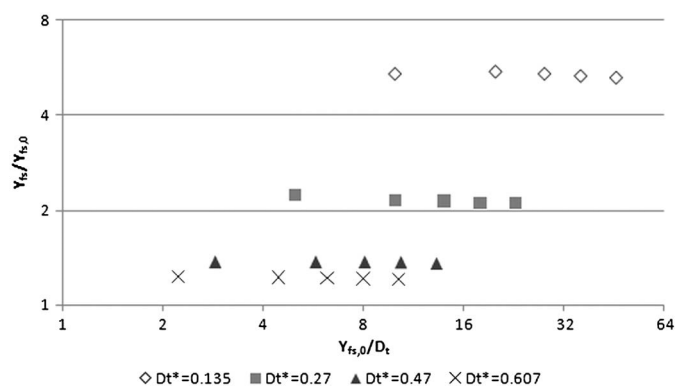


Fig. 12. Numerical predictions of normalized free-surface level rise for different initial free-surface levels and ventilation tower diameters

a larger-diameter tower has a longer perimeter that increases the film flow generated as the air pocket moves upward. One needs to consider, however, that other geometry characteristics of the ventilation tower, such as the length, have an important effect in determining whether geysering will occur as air pockets are released.

The other parameters considered in the investigation—initial air phase pressure and initial water level in the ventilation tower—had secondary importance to the occurrence of geysering generated by a large air-pocket release. The initial air-phase pressure head and the initial water level in the ventilation tower basically determined the initial free-surface level in the ventilation tower as the air pocket reached the base of the ventilation tower. As expected, higher water levels were more likely to result in geysering as the air pocket moved upward.

The predictions of the trajectory of the free-surface level and the air-water interface level variation agreed reasonably well with the experimental measurements. On the other hand, the predictions for the other modeled variables were not so accurate, even though important flow features were captured. The oscillations observed in the measurements for the largest ventilation tower were reproduced, although the amplitude and frequency were not correct. The predictions for the smallest ventilation tower also agreed with the experiments and predicted no oscillations but a monotonic increase in the air-water interface level and velocities and decrease in the pressure. The numerical model indicated that the upward motion of the water column can be accelerated by the geyser occurrence because the spilled water causes a further decrease to the column weight.

One consideration is how the results obtained with these experiments and the numerical model could be scaled to full-scale ventilation towers with presumably turbulent film flows. Not only would the film thickness vary with time and space, but there is also the possibility of shear flow instabilities between the air pocket and the film. There are likely to be other effects that do not scale properly between the laboratory experiments and field events at a much larger scale, so the current investigation should be considered as a first attempt at understanding the phenomena involved.

Conclusions

The experiments achieved the general objective to reproduce and characterize geysering events generated by the release of large air pockets through water-filled ventilation towers. The experimental variables considered in the investigation were the initial water level in the ventilation tower, the initial air-phase pressure, and the diameter of the ventilation tower. Thirty-six different configurations of these variables were tested.

This work represents an initial step, as there is essentially no guidance on how to avoid geysers created by large pockets of entrapped air. To date, most investigations focused on inertial oscillations as the mechanism for geysering, whereas few experimental studies explored geysering caused by the release of air pockets. This probably explains why there appears to be no widespread understanding of the importance of air release during the tunnel-filling process.

Results show that the diameter of the ventilation tower is a critical variable. Although there needs to be further investigations, this is an important finding. Other geometry characteristics, such as the tower length, also have an important effect in determining whether geysering will occur. More investigation, preferably in larger-scale apparatus, is required to determine how other factors, such as shear flow instabilities, affect the air release through such towers.

A numerical model was constructed to simulate the experimental results. The approach used was to model the rise of the water column in the ventilation tower by using a lumped inertia approach. The upward motion of the water column was caused by the air-phase pressure and countered by the weight of the water column and friction. The length of the water column decreased over time because of the film flow, which was assumed to be independent of the air pressure-driven motion of the water column. Although simplifying assumptions were introduced in the model development, the predictions of the free-surface level and the air-water interface level variation over time agreed reasonably well with the experimental measurements. On the other hand, the predictions for the other modeled variables were not so accurate, although some important flow features were qualitatively predicted. The numerical model indicated that the upward motion of the water column can be

accelerated by the geyser occurrence, because the spilled water causes a further decrease to the column weight.

Although the experimental investigation and the numerical model presented in this study are regarded as a significant step in comprehending geysering occurrences in drainage systems, there are several other important issues that still need to be addressed in further investigations. One of these issues is the numerical description of the motion of entrapped air pockets in the sewers, which could help determine at which points these pockets will be ventilated. A second issue is to perform experiments on geysering caused by the release of large air pockets at larger scales to assess the effect of turbulent film flows and the possible shear flow instabilities between the film and the air flow.

Notation

The following symbols are used in this paper:

- A = ventilation tower cross-sectional area;
- C_a = constant equal to P_{a0}/ρ_a^l ;
- D = pipe or valve diameter;
- D_t = ventilation tower diameter;
- D_t^* = nondimensional ventilation tower diameter = D_t/D ;
- f = friction coefficient;
- g = gravity acceleration;
- H = water pressure head;
- H_a = air pressure head;
- H_{atm} = atmospheric pressure head;
- H^* = nondimensional water pressure head = H/L ;
- L = length of ventilation tower;
- P_{a0} = initial absolute air pressure before butterfly valve opening;
- $Q_{displaced}$ = upward water flow in ventilation tower;
- Q_f = film flow;
- T_{ref} = reference time after valve opening;
- T_{ref}^* = nondimensional reference time after valve opening = $T_{ref}/(L/\sqrt{gD_t})$;
- t = time;
- U_∞ = rising velocity of Taylor bubble;
- V_a = air-phase volume;
- V_{fs} = free-surface velocity;
- V_{fs}^* = nondimensional free-surface velocity = $V_{fs}/\sqrt{gD_t}$;
- V_{int} = air-water interface velocity;
- V_{int}^* = nondimensional air-water interface velocity = $V_{int}/\sqrt{gD_t}$;
- Y_{int} = air-water interface coordinate;
- Y_{int}^* = nondimensional air-water interface coordinate = Y_{int}/L ;
- Y_{fs} = free-surface coordinate;
- Y_{fs}^* = nondimensional free-surface coordinate = Y_{fs}/L ;
- γ = polytropic coefficient;
- δ = film flow thickness;
- ϵ = pipe material roughness;
- ρ_a = air-phase density; and
- ρ_w = water-phase density;

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